General Chemistry (InCh1006)

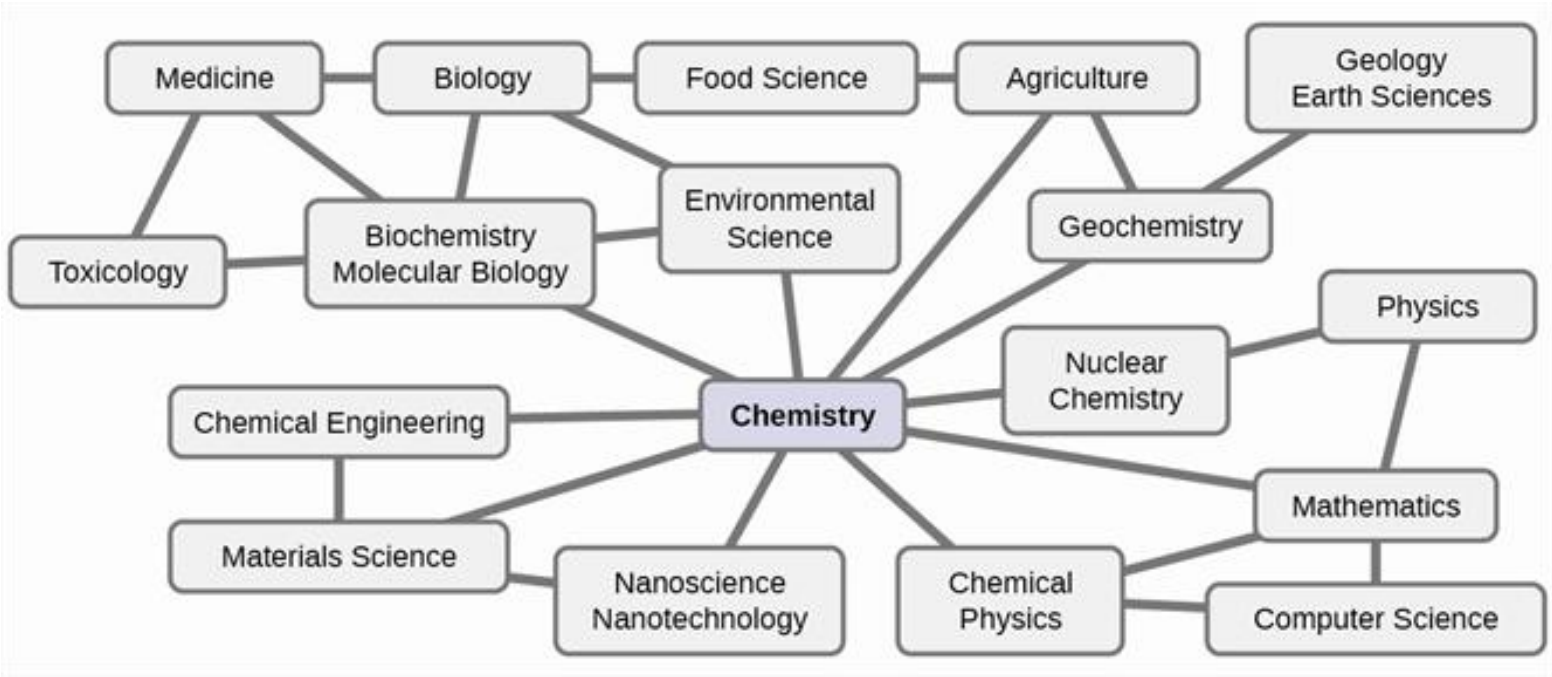
Chapter 1

Properties, Measurements and Units

1.1. Introduction

Chemistry: The study of the composition, properties, and interactions of matter.

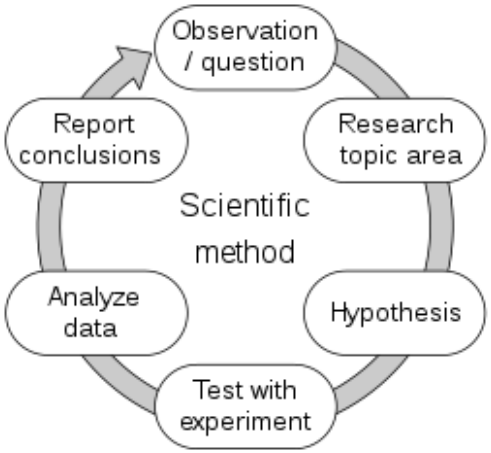
- **Chemistry** is interconnected with a vast array of science, technology, engineering, and math fields



The Scientific Method

- **Chemistry** is a science based on observation and experimentation.

➢ It involves attempting to answer questions and explain observations in terms of the laws and theories of chemistry, using procedures that are accepted by the scientific community – **The Scientific Method**



➢ There is no single route to answering a question or explaining an observation, but there is an aspect common to every approach:

- ✓ Each uses knowledge based on experiments that can be reproduced to verify the results.
- **Hypothesis:** A tentative explanation of observations that acts as a guide for gathering and checking information.
- ✓ It is tested by experimentation, calculation, and/or comparison with the experiments of others and then refined as needed.
- **Laws (**of science**):** It summarizes a vast number of experimental observations, and describe or predict some aspect of the natural world.
- A scientific law focuses solely on describing **what**.
- A scientific law provides a description of a directly observable phenomenon. It describes what will or is expected to happen in a certain set of circumstances.

➢If such a hypothesis turns out to be capable of explaining a large body of experimental data, it can reach the status of a theory.

▪**Theory:** A well-substantiated, comprehensive, testable explanations of particular aspects of nature.

✓can be modified if new data become available

✓What is a scientific theory explores **why**. A theory is about underlying causes, seeking to explain the reason the phenomenon occurs.

✓The focus of a theory is to provide a logical explanation for things that occur in nature. There can be more than one theory about the same phenomenon.

- **Scientific method:** The path of discovery that leads from **question** and **observation** to **law** or **hypothesis** to **theory**, combined with experimental verification of the hypothesis and any necessary modification of the theory

The Domains of Chemistry

- Chemists study and describe the behavior of matter and energy in three different domains: macroscopic, microscopic, and symbolic.

1. **Macroscopic domain:** Large enough to be sensed directly by human sight or touch
2. **Microscopic domain:** Often pictured in the mind such as atoms, ions, electrons, protons, neutrons, chemical bonds, etc.

✓ Some are visible through a microscope, such as viruses

3. **Symbolic domain:** Contains the specialized language used to represent components of the macroscopic and microscopic domains such as chemical symbols, chemical formulas and chemical equations

Phases and Classification of Matter:

Matter is defined as anything that **occupies space and has mass**.

Phases of matter: Matter can exist in four phases (states).

1. **Solid:** It is rigid and possesses a definite shape and volume.
2. **Liquid:** It flows, possesses volume and takes the shape of a container
3. **Gas:** It takes both the shape and volume of its container.

▪Solid and liquid samples have volumes that are very nearly independent of pressure.

4. Plasma: A gaseous state of matter that contains appreciable numbers of electrically charged particles

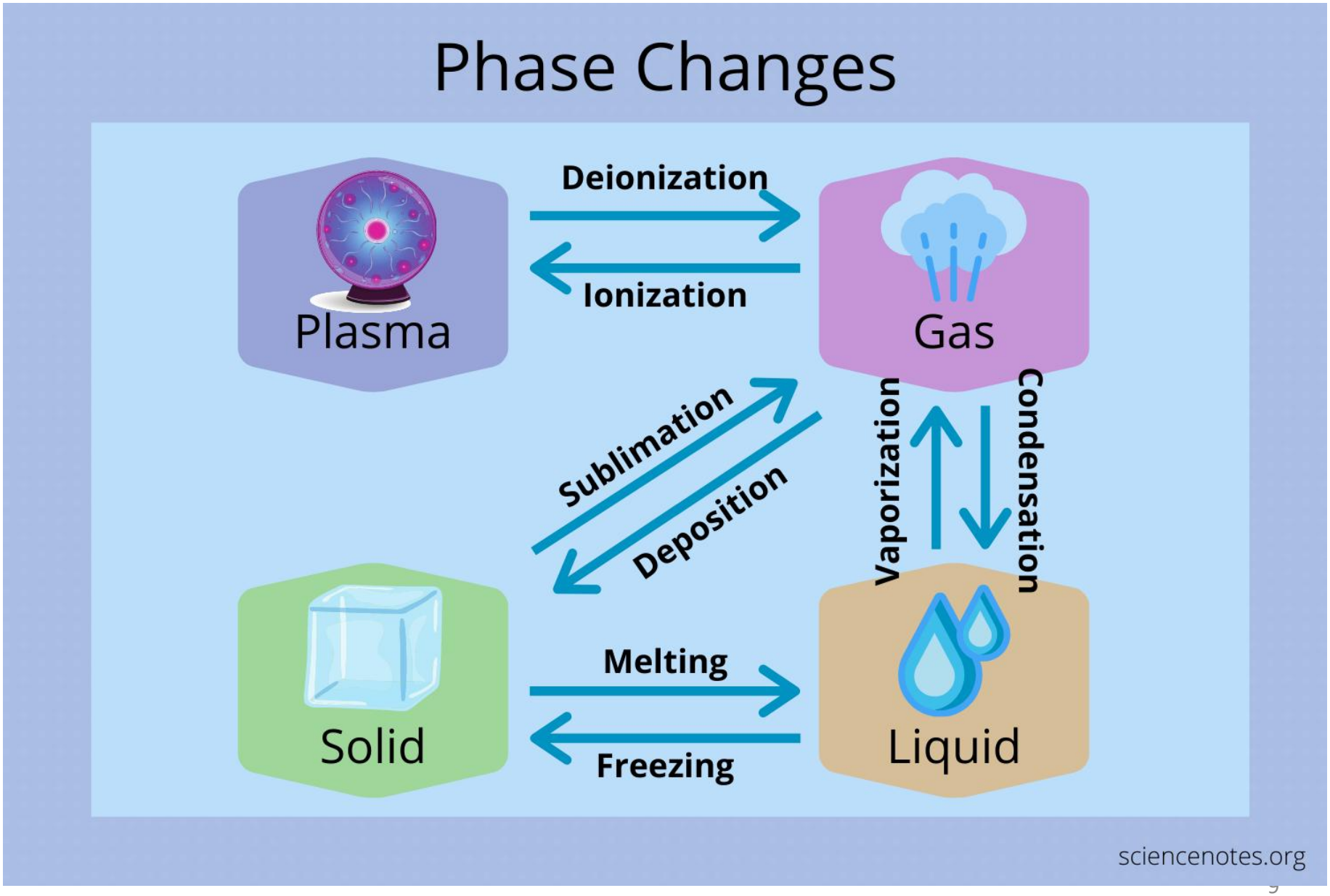
▪It is found in

✓stars,

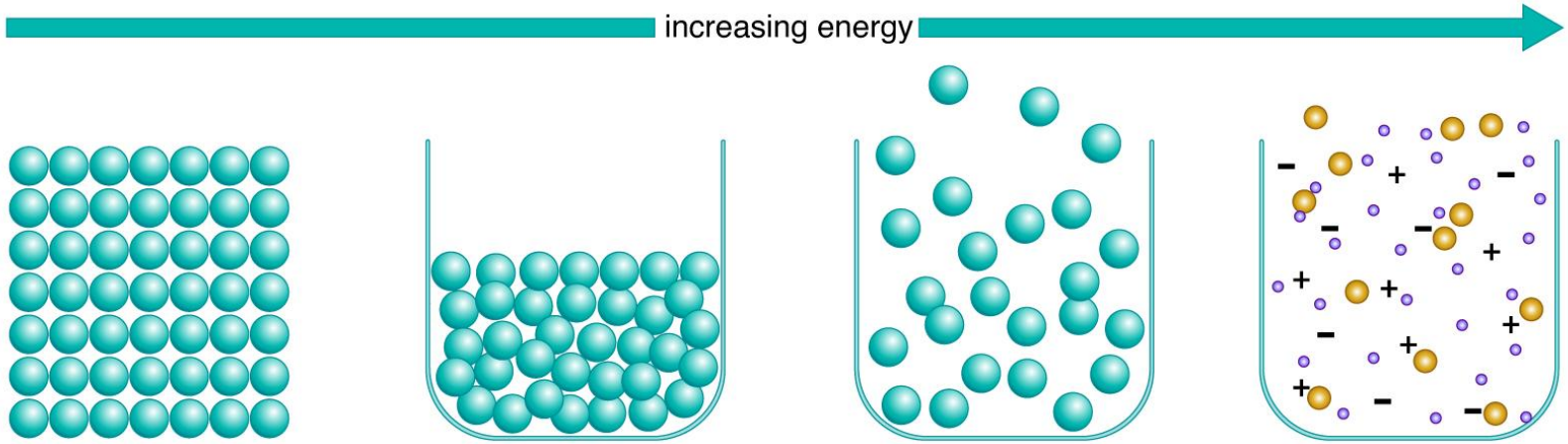
✓high-temperature environments such as lightning strikes,

✓certain television screens and specialized analytical instruments

Transformation of Phases



The Four phases (states) of matter.



Solid

The molecules that make up a solid are arranged in regular, repeating patterns. They are held firmly in place but can vibrate within a limited area.

Liquid

The molecules that make up a liquid flow easily around one another. They are kept from flying apart by attractive forces between them. Liquids assume the shape of their containers.

Gas

The molecules that make up a gas fly in all directions at great speeds. They are so far apart that the attractive forces between them are insignificant.

Plasma

At the very high temperatures of stars, atoms lose their electrons. The mixture of electrons and nuclei that results is the plasma state of matter.

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Classifying Matter

A) Pure substance: It has a constant composition.

- All specimens of a pure substance have exactly the same makeup and properties.

E.g.: Any sample of sucrose (table sugar) consists of 42.1% carbon, 6.5% hydrogen, and 51.4% oxygen by mass.

- Pure substances are divide into two classes:

1. Elements: Pure substances that cannot be broken down into simpler substances by chemical changes

Examples: Iron, silver, sulfur, oxygen

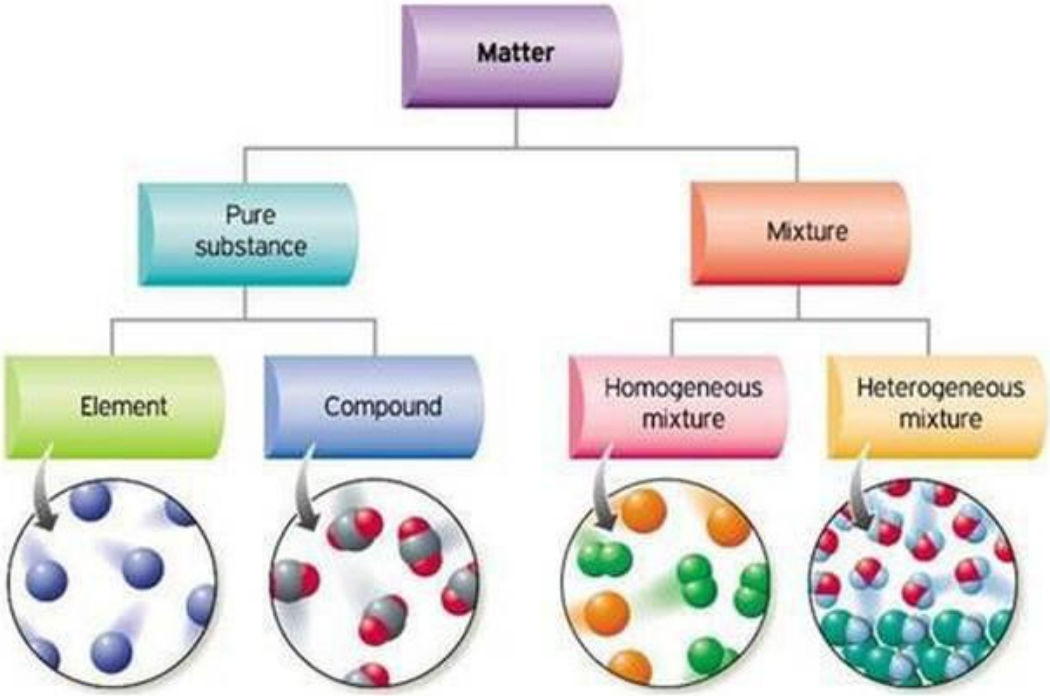
2. Compounds: Pure substances that can be broken down by chemical changes

- ➢ The breakdown may produce either elements, other compounds, or both.
- ✓ Mercury(II) oxide, can be broken down by heat into the elements mercury and oxygen .
- ✓ When heated in the absence of air, the compound sucrose is broken down into the element carbon and the compound water.

B) Mixture : It is composed of two or more types of matter that can be present in varying amounts and can be separated by physical changes, such as evaporation

1. Heterogeneous mixture: A mixture with a composition that varies from point to point.

2. Homogeneous mixture (solution): Exhibits a uniform composition and appears visually the same throughout.



1.3. The Properties of Substances: Physical and Chemical Properties

Properties: The characteristics that enable us to distinguish one substance from another.

A. **Physical property :** It is a change in the state or properties of matter without any accompanying change in its chemical composition

- It is a characteristic of matter.

Examples: density, color, hardness, melting and boiling points, electrical conductivity, etc.

- We can observe some physical properties, such as density and color, without changing the physical state of the matter observed.
- Physical properties, such as melting temperature or freezing temperature, can only be observed as the matter undergoes a physical change.

B. Chemical property: It is the change of one type of matter into another type (or the inability to change).

Examples: flammability, toxicity, acidity, reactivity, etc.

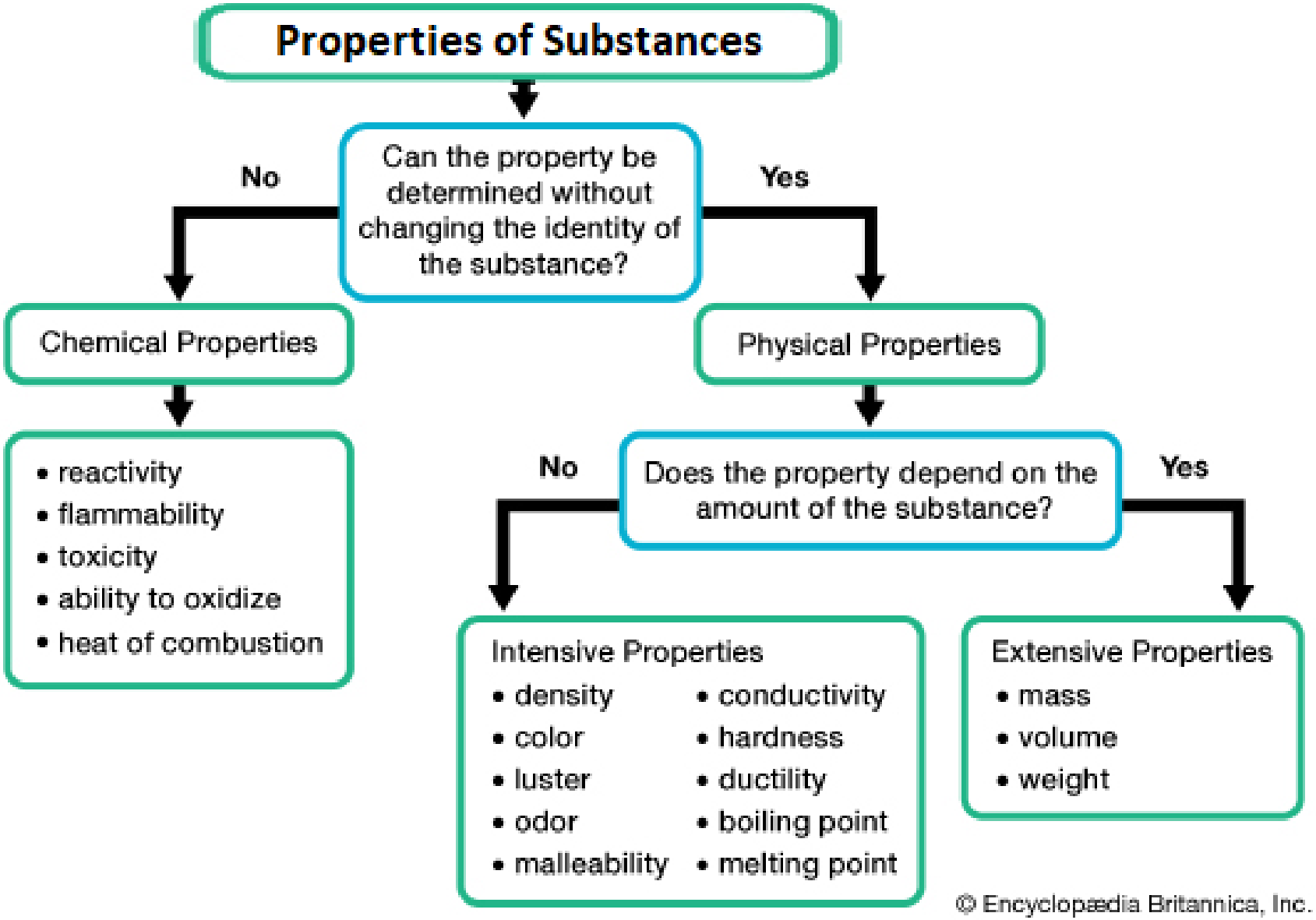
- Iron combines with oxygen in the presence of water to form rust while chromium does not.

C. Extensive property: A property that depends on the amount of matter present

Examples: The mass and volume of a substance, heat

D. Intensive property: The property of a sample of matter that does not depend on the amount of matter present.

Example: Temperature, density, boiling and melting point, conductivity etc.


Properties of Substances
Can the property be determined without changing the identity of the substance?
No
Yes
Chemical Properties
• reactivity
• flammability
• toxicity
• ability to oxidize
• heat of combustion
Physical Properties
Does the property depend on the amount of the substance?
No
Yes
Intensive Properties
• density
• color
• luster
• odor
• malleability
• conductivity
• hardness
• ductility
• boiling point
• melting point
Extensive Properties
• mass
• volume
• weight
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1.4 Measurements and Units

- Measurements provide the macroscopic information that is the basis of most of the hypotheses, theories, and laws that describe the behavior of matter and energy in both the macroscopic and microscopic domains of chemistry.
- Every measurement provides three kinds of information:

➢ the size or magnitude of the measurement (a number)

➢ a standard of comparison for the measurement (a unit) and

➢ an indication of the uncertainty of the measurement

- The number in the measurement can be represented in decimal form and scientific (exponential) notation.

Example: The mass of an object is :

= 0.0000025 kilograms (decimal form)

$= 2.5 \times 10^{-6}$ kg (scientific notation)

1.4.1. The International System of Units (SI Units)

Units: They are standards of comparison for measurements

- The standards for these units are fixed by international agreement, and they are called the **International System of Units** or **SI Units**

Base Units of the SI System

Property Measured	Name of Unit	Symbol of Unit
length	meter	m
mass	kilogram	kg
time	second	s
temperature	kelvin	K
electric current	ampere	A
amount of substance	mole	mol
luminous intensity	candela	cd

Sometimes we use units that are fractions or multiples of a base

Common Unit Prefixes

Prefix	Symbol	Factor
femto	f	10^{-15}
pico	p	10^{-12}
nano	n	10^{-9}
micro	μ	10^{-6}
milli	m	10^{-3}
centi	c	10^{-2}
deci	d	10^{-1}

Prefix	Symbol	Factor
kilo	k	10^{3}
mega	M	10^{6}
giga	G	10^{9}
tera	T	10^{12}

Derived SI Units

We can derive many units from the seven SI base units.

Volume: The measure of the amount of space occupied by an object. The standard SI unit of volume is defined by the base unit of length The standard volume is a **cubic meter (m^3)**

- A cube with edge lengths of one decimeter contains a volume of one cubic decimeter (dm^3).

  1 **liter (L) = 1dm^3**

  1 milliliter (mL) = 1 cubic centimeter (cm^3 = cc)

Density: The density of a substance is the ratio of the mass of a sample of the substance to its volume.

- The units of density are defined by the base units of mass and length.
- SI unit for density: kilogram per cubic meter (kg/m^3).
- More convenient unit:
- grams per cubic centimeter (g/cm^3) – for solids and liquids
- grams per liter (g/L) - for gases

1.4.2. Measurement: Uncertainty, Accuracy, and Precision

- Counting is the only type of measurement that is free from uncertainty.
- ➢ The result of such a counting measurement is an example of an **exact number**.
- The numbers of measured quantities, unlike directly counted quantities, are not exact.

Uncertainty: Every measurement has some uncertainty, which depends on the device used and/or the user's ability.

Example: Suppose on a standard electronic balance, you obtained a reading of 6.72 g for an object,

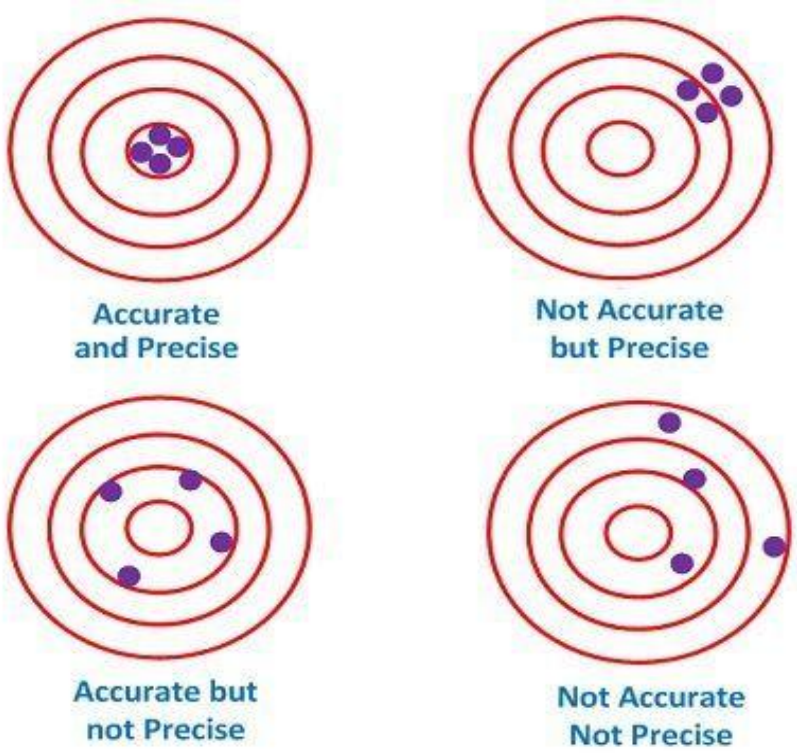
- ✓ the digits 6 and 7 are certain, and
- ✓ the 2 indicates that the mass of the object is likely between 6.71 and 6.73 grams.
- ➢ The object weighs about 6.72 grams, with a nominal uncertainty in the measurement of **± 0.01 gram**.

Precision: Measurements are said to be precise if they yield very similar results when repeated in the same manner.

- Precise values agree with each other

Accuracy: A measurement is considered accurate if it yields a result that is very close to the true or accepted value.

- Accurate values agree with a true value



Example:

- Suppose a quality control chemist at a pharmaceutical company is tasked with checking the accuracy and precision of three different dispenser machines that are meant to dispense 296 mL of cough syrup into storage bottles.
- Procedure: From each machine fill, say five bottles and then determine the actual volumes dispensed

Results:

Dispenser 1	Dispenser 2	Dispenser 3
283.8	298.3	296.1
284.1	294.2	295.9
283.9	296.0	296.1
284.0	297.8	296.0
284.1	293.9	296.1

Dispenser	Finding
1	Precise but not accurate
2	more accurate but less precise
3	most accurate and precise

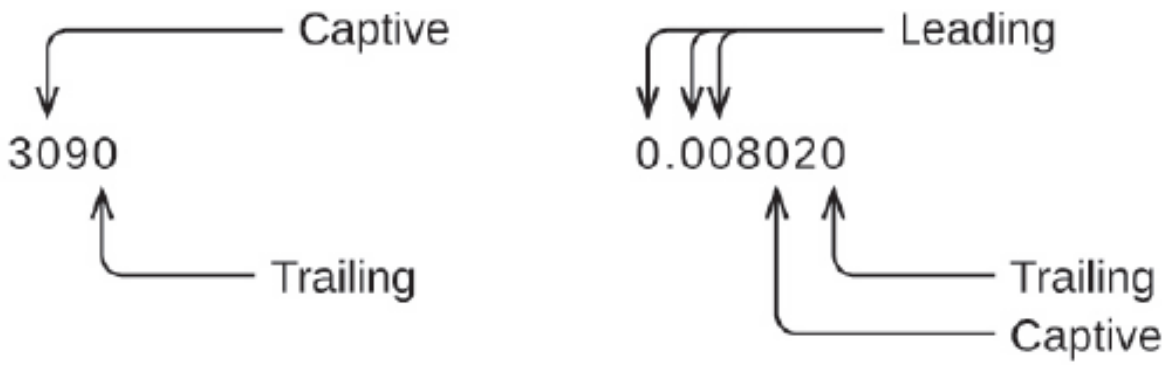
1.4.3. Significant figures:

- All of the digits in a measurement, including the uncertain last digit, are called significant figures or significant digits
- **Whenever you make a measurement properly, all the digits in the result are significant.**
- **If you were analyzing a reported value,**

✓ all nonzero digits may not be significant

✓ The zeros require some thought.

- To deal with the zeros, the terms "leading," "trailing," and "captive" zeros have to be considered



- ✓ Captive zeros result from measurement and are therefore always significant
- ✓ Leading zeros are never significant - they tell us where the decimal point is located.
- ✓ The number of significant figures is uncertain in a number that ends with a zero to the left of the decimal point location.

Examples:

1356 m – number of significant figures = 4

34.0 g – number of significant figures = 3

80.406 mL – number of significant figures = 5

0.002703 L = 2.703×10^{-3} – number of significant figures = 4

(0.00 or 10^{-3} locates the decimal point)

2300 g – the zeros may be significant or they could simply indicate where the decimal point is located

- ✓ The ambiguity can be resolved with the use of exponential notation
- ➢ 2.3×10^{3} - number of significant figures = 2
- ➢ 2.30×10^{3} - number of significant figures = 3
- ➢ If only the decimal-formatted number is available, better to assume that all trailing zeros are not significant.

Significant Figures in Calculations

- Results calculated from a measurement are at least as uncertain as the measurement itself.

➢ We must take the uncertainty in our measurements into account to avoid misrepresenting the uncertainty in calculated results.

✓ One way to do this is to report the result of a calculation with the correct number of significant figures

Three rules for rounding numbers:

1. When we add or subtract numbers, we should round the result to the same number of decimal places as the number with the least number of decimal places (the least precise value in terms of addition and subtraction).

2. When we multiply or divide numbers, we should round the result to the same number of digits as the number with the least number of significant figures (the least precise value in terms of multiplication and division).

3. If the digit to be dropped (the one immediately to the right of the digit to be retained) is

- **less than 5**, we "round down" and leave the retained digit unchanged;
- **more than 5**, we "round up" and increase the retained digit by 1
- **5**, we round up or down, whichever yields an even value for the retained digit.

Examples to round to three significant figures:

- 0.028675 rounds "up" to 0.0287 (the dropped digit, 7, is greater than 5)
- 18.3384 rounds "down" to 18.3 (the dropped digit, 3, is less than 5)
- 6.8752 rounds "up" to 6.88 (the dropped digit is 5, and the retained digit is odd)
- 92.85 rounds "down" to 92.8 (the dropped digit is 5, and the retained digit is even)

Examples for addition and subtraction with significant figures:

a) Add 2.024 g and 4.38 g.

$$2.024\ g + 4.38\ g = 6.404\ g$$

Round to two decimal points.

Answer: 6.40 g

b) Subtract 124.23 g from 581 g.

581 g – 124.23 g = 456.77 g

Round to zero decimal point

Answer: 457

Examples for multiplication and division with significant figures:

a) Multiply 0.624 cm by 5.8 cm.

0.624 cm x 5.8 cm = 3.6192 cm^2

Round to two significant figures

Answer: 3.6 cm^2

b) Divide 87.171 g by 24.5 mL

87.171 g ÷ 24.5 mL = 3.558 g/mL

Round to three significant figures

Answer : 3.56 g/mL

Mathematical Treatment of the Units of Measurement

Dimensional analysis (factor - label method): The units of quantities must be subjected to the same mathematical operations as their associated numbers.

Example

Calculate the time required for a person running at a speed of 10m/s to travel a distance of 25 m.

Answer:

$$time = \frac{distance}{speed} = \frac{25\ m}{10m/s} = 2.5\ s$$

- Note that the units are also divided ($\frac{m}{m} \times s = s$)

Conversion Factors and Dimensional Analysis

- A ratio of two equivalent quantities expressed with different measurement units can be used as a **unit conversion factor**.

Example:

The masses of 1 lb and 453.59 g are equivalent

The unit conversion factor is derived from the ratio,

$$\frac{453.59\ g}{1\ lb} = 453.59\ \frac{g}{lb}$$

Example: What is the mass of 3 lb of an object in gram?

Answer: $3\text{lb} \times 453.59\,\frac{g}{lb} = 1360.77g$

Conversion of Temperature Units

Celsius scale: 0 °C is defined as the freezing temperature of water and 100 °C as the boiling temperature of water

Fahrenheit scale: The freezing point of water is defined as 32 °F and the boiling temperature as 212 °F.

The equation relating the temperature scales is:

$$T(^oF) = \frac{9}{5} \times T(^oC) + 32$$

Kelvin scale: The freezing temperature of water on this scale is 273.15 K and its boiling temperature 373.15 K.

$$T\ (K) = T(^oC) + 273.15$$